DEVESH BHARDWAJ

Data Engineer | Backend Engineer | GenAI Developer | Solution Architect
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PROFESSIONAL SUMMARY

Software engineer with 4 years at Tata Consultancy Services building cloud-native data, backend and AI systems for global life sciences clients. Delivered AWS ETL pipelines processing 1M+ records daily from 8 sources, architected an event-driven access-management platform on FastAPI, and shipped NLP solutions cutting manual effort by up to 85%.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, FastAPI, REST API Design, Pydantic, Asyncio, Stored Procedures, Pandas
AWS: Glue, Glue Crawlers, Glue Data Catalog, AppFlow, Redshift, Aurora PostgreSQL, DynamoDB, SQS, EventBridge, S3, IAM, CloudWatch
Deployment: Github, Github Actions, Cloudformation, Docker, Kubernetes
Azure: Bot Framework Composer, App Service, Azure Functions, Blob Storage, Conversational Language Understanding (CLU/LUIS)
Data Engineering: ETL/ELT Design, Apache Airflow DAGs, Data Lake & Data Modeling, Incremental Loads, Idempotent Upserts, Schema Evolution
AI / ML / NLP: NLP, Multi-class Text Classification, Hugging Face Sentence-Transformers, Vector Embeddings, Intent & Entity Recognition, Conversational AI, RAG, LLMs, Langchain, LangGraph
Architecture & DevOps: Microservices, Event-Driven & Serverless Architecture, Message Queues (SQS/DLQ), RBAC, GitHub Actions CI/CD, Agile/Scrum
Domain & Platforms: Veeva CRM, Veeva RIM, Veeva CDMS, ServiceNow, Workday, LMS Integration, Life Sciences / GxP, Jira

PROFESSIONAL EXPERIENCE

Tata Consultancy Services | August 2022 – Present | Noida,Uttar Pradesh

Solution Architect & Backend Engineer - Veeva RIM Access Management System | December 2024 – Present

- Architected a microservices-based platform automating access provisioning, modification and de-provisioning for Veeva RIM, serving 10000+ users and cutting request turnaround from 2 days to ~1 hours.
- Built asynchronous FastAPI REST services with Pydantic validation and role-based access control, powering request creation, an admin configuration panel and approval workflows with e-signature and audit trail.
- Designed a centralized, template-driven email notification service handling 40+ event types, removing duplicated logic across 5 modules and cutting new-notification build time by 98%.
- Automated training-compliance validation via REST integrations with the Learning Management System, Workday and ServiceNow, eliminating 100% of non-compliant grants.
- Engineered an event-driven backend on Amazon SQS with dead-letter queues, EventBridge schedulers, Aurora and DynamoDB session management, sustaining 3000+ requests/month.

GenAI Developer - Conversational Form-Filling Assistant | March 2023 – November 2024

- Developed a card-based conversational assistant on Azure Bot Framework Composer that auto-drafts a 70-field business form from free-text input, cutting completion time from 10–15 minutes to under 2 (~85% reduction) for 500+ users.
- Implemented intent classification and entity extraction with Azure Language Understanding, paired with a Hugging Face Sentence-Transformers embedding model and cosine-similarity semantic field mapping at 90% accuracy.
- Deployed a serverless stack on Azure App Service and Functions, with Blob Storage for conversation state and transcript persistence.

Machine Learning Engineer - Automated Incident Triage Tool | January 2023 – April 2023

- Trained and deployed a supervised multi-class NLP classification model that auto-routes ServiceNow incidents to the correct assignment group at 80% accuracy on 5000+ historical tickets.
- Automated triage of 1000+ incidents/month via a scheduled batch inference job, correcting misrouted tickets and reducing mean time to resolution by 50%.

Data Engineer - Veeva CRM Data Pipeline | August 2022 – January 2023

- Engineered a production ETL pipeline ingesting Veeva CRM data from 8 sources via AWS AppFlow, transforming and upserting 1M+ records (~5 GB) daily into a Redshift data lake for 5+ downstream consumers.
- Orchestrated dependency-aware Apache Airflow DAGs over AWS Glue jobs, Crawlers and Data Catalog, automating schema evolution, incremental loads, retries and backfills.
- Built fault detection auto-raising ServiceNow incidents on Glue job failure, cutting mean time to detect from 4-5 hours to under 5 minutes.
- Authored Redshift stored procedures for idempotent staging-to-target upserts plus DynamoDB run-level audit logging; automated multi-environment deployment with GitHub Actions CI/CD.

EDUCATION & CERTIFICATIONS

B.Tech in Computer Science - Himachal Pradesh University, Shimla | 2022 | 78%
Certifications: AWS Certified Solutions Architect